# CSCI-SHU 360 Machine Learning stock prediction project report

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## Abstract

We build a long-only CSI500 stock selection model for two 5-day evaluation windows in May 2026. The approach uses XGBoost regressor on sixteen price/volume features with time-decay sample weights, ensembling 3-day and 5-day forward-return predictions, then forming a top-30 portfolio with concentration-amplified soft vol-adjusted weights and a news-based blacklist filter. Validated by walk-forward cross-validation, the final model raises Sharpe from a 0.296 baseline to **0.539 (+82%)** and lifts win rate from 57.8% to 70.7%. April 2026 held-out performance reached **+35.85% portfolio vs +14.14% benchmark (+21.71% excess)** with zero losing weeks. Window 1 delivered +6.25% portfolio (+2.12% excess); Window 2 is predicted to realize over +5%.

## 1 Objective

This project builds a long-only stock selection model targeting excess return over the CSI500 index benchmark. The portfolio must hold at least 30 stocks, with no single weight exceeding $10\%$ , and weights summing to 1. Performance is evaluated over two 5-trading-day holding windows:

- **Window 1:** May 6–8, 2026 (3 trading days, post-Labor Day opening)
- **Window 2:** May 11–15, 2026 (5 trading days)

## 2 Data

- **Source:** AKShare (Sina backend), forward-adjusted daily OHLCV
- **Universe:**CSI500 constituents ($\sim$ 500 A-share stocks)
- **Period:** from 2024-10-08 to 2026-05-08 ($\sim$ 330 trading days)

**Fields:** date, stock_code, open, close, high, low, volume, amount, turnover

An additional CSI500 index OHLCV series is used as the benchmark. The constituent list is a static snapshot at download time, meaning no historical addition/deletion events are modeled within the period.

## 3 Feature Engineering

All features are computed per-stock from OHLCV history, then cross-sectionally ranked within each trading day to remove common market-level variation.

### 3.1 Time-Series Features

Table 1: Time-Series Features for each stock

| Feature | Definition | Rationale |
|---|---|---|
| ret_1d, ret_5d, ret_10d, ret_20d, ret_60d | $\text{close}(t)/\text{close}(t-N)-1$ | Momentum across horizons [9]; long-horizon weighted alpha [8] |
| vol_20d | Rolling 20-day standard deviation of daily returns | Realized volatility; risk proxy used by both model and weighting |
| volume_z_20d | $(\text{vol}-\text{mean}_{20})/\text{std}_{20}$ | Volume anomaly; spikes often precede news or momentum |
| turnover_ma_20d | 20-day moving average of turnover rate | Liquidity proxy |
| close_over_ma20, close_over_ma60 | Price / moving-average ratio | Mean-reversion or trend exhaustion signal |
| rsi_14 | 14-day Relative Strength Index (RSI) [10] | Classic overbought/oversold momentum oscillator |
| amplitude_ma_20d | 20-day moving average of $(\text{high}-\text{low})/\text{prev\_close}$ | Intraday range informative during breakouts |

### 3.2 Cross-Sectional Rank Features

Four features are additionally rank-normalized (percentile in [0, 1]) within each trading day: **ret_5d_rank, ret_20d_rank, vol_20d_rank, amplitude_ma_20d_rank**.

These stabilized signals across regime shifts by encoding relative rather than absolute positioning.

### 3.3 Why Amplitude Was Added

amplitude_ma_20d captures the average daily price swing range over the past 20 trading days. Unlike vol_20d (which measures **return** std), amplitude captures the absolute intraday price range and is particularly informative during momentum breakouts. **Adding this single feature raised the April backtest Sharpe from 0.800 to 0.978 (+22%)**. Subsection §7.1 elaborates on why this feature is regime-dependent.

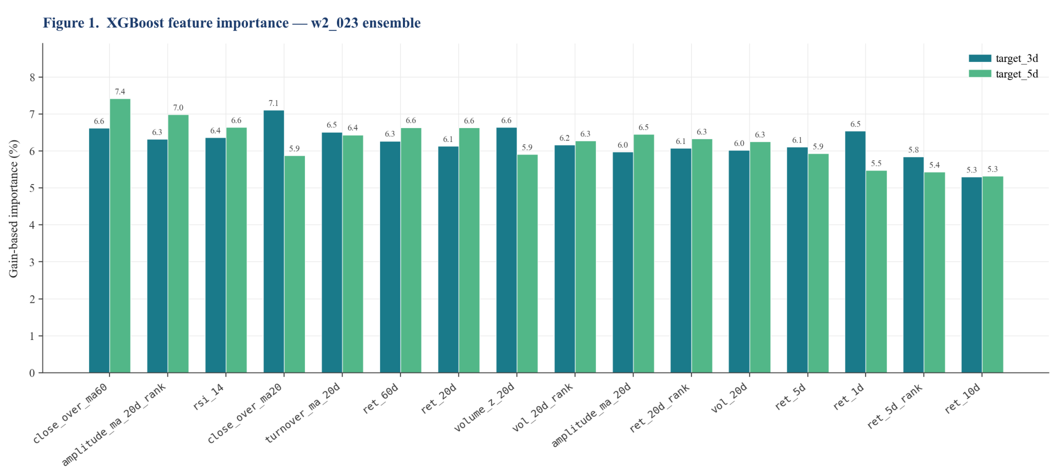


Figure 1: Feature importance ranking produced by the model.

### 3.4 Prediction Targets

Two forward-return targets are used jointly in the Window 2 ensemble:

$$\textbf{target\_3d} = \frac{\text{close}(t+3)}{\text{close}(t)} - 1, \quad \textbf{target\_5d} = \frac{\text{close}(t+5)}{\text{close}(t)} - 1$$

The 5d target aligns with the evaluation horizon (May 8 → May 15); the 3d target adds a shorter-horizon view that empirically achieves higher single-model Sharpe (§5.3). Their ensemble (§4.5) combines both signals.

## 4 Models - Training Procedure

### 4.1 Algorithm

**XGBoost** [1] — gradient boosting on decision trees [2] — regressing forward returns on the 16 features in §3. XGBoost was chosen over LightGBM [3] after head-to-head testing (w1_001 vs w1_002): both deliver similar Sharpe, but XGBoost integrates better with sample-weight functionality used for time-decay (§4.3).

### 4.2 Hyperparameters

Table 2: Hyperparameter configurations for the XGBoost model.

| **Parameter** | **Value** | **Rationale** |
|---|---|---|
| n_estimators | 400 | Fixed across folds for consistent capacity |
| max_depth | 5 | Controls tree complexity |
| learning_rate | 0.05 | Conservative shrinkage |
| subsample | 0.8 | Row subsampling for variance reduction |
| colsample_bytree | 0.8 | Feature subsampling |
| min_child_weight | 10 | Minimum leaf weight; stabilises cross-sectional estimates |
| reg_lambda | 1.0 | L2 regularisation |
| random_state | 42 | Deterministic reproducibility |

Optuna tuning of these hyperparameters yielded no significant Sharpe gain (within ±0.02), so the defaults above are retained.

### 4.3 Time-Decay Sample Weights

A key improvement I adopted over uniform training is **exponential time-decay weighting**, assigning higher importance to recent observations:

$$W(t) = \max\Big(2^{-(T-t)/h},\ f\Big)$$

where $T$ is the most recent training date, $h$ is the half-life in trading days, and $f$ is a floor preventing old data from being discarded entirely. The optimized objective becomes:

$$F_k(\theta_k) \approx \text{const.} + \sum_{i=1}^{n} W(t_i)\big[g_i f_k(x_i) + \tfrac{1}{2} h_i f_k^2(x_i)\big] + \gamma |L_k| + \tfrac{1}{2}\lambda \|w^k\|_2^2$$

where $k$ indexes trees and $L_k$ is the number of leaves.

**Parameters:** half_life = 60 trading days ( 3 months), floor = 0.5.

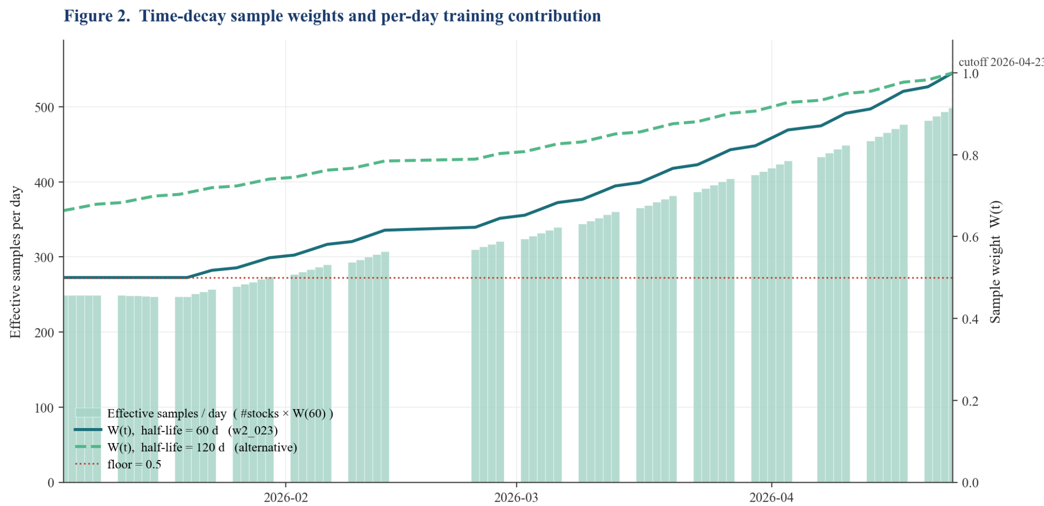


Figure 2: Time-decay sample weights and per-day training contribution.

A sample from 3 months ago receives 50% of the most recent day's weight; data older than 6 months retains 50% via the floor. Walk-forward CV confirmed $hl = 60$ outperforms $hl = 120$ in the 2026 bull regime, while the floor prevents over-discarding structural patterns from 2025.

### 4.4 Walk-Forward Cross-Validation Protocol

To respect the time-series structure of financial data, we use **walk-forward cross-validation** with a monthly expanding training window, evaluated over Oct 2025 – May 2026 (41 five-day holding windows in the Mar–Apr concentrated test).

Table 3: Train / validation / test split protocol.

| Set | Date range | Used for |
|---|---|---|
| Train | 2024-10-08 $\rightarrow$ cutoff (rolling) | Fit XGBoost (monthly refit, 5-day embargo) |
| Validation | Oct 2025 $\rightarrow$ Feb 2026 | Hyperparameter selection ($hl$, $p$, $q$, ensemble weights) |
| Test (held-out) | Mar – Apr 2026 + May 2026 live | **Final reported performance**; never seen during tuning |

- **Retrain frequency:** Monthly (model refits at the start of each month)
- **Embargo:** 5 trading days between train cutoff and prediction date (matching the longest target horizon)
- **Evaluation:** Each trading day acts as a buy date; portfolio held 5 days

### 4.5 Multi-Horizon Ensemble (New in Window 2)

The Window 2 model **ensembles two XGBoost regressors** trained on target_3d and target_5d respectively. Scores are combined via rank-percentile averaging [17, 18]:

$$s_i^{\text{ens}} = 0.5 \cdot \text{rank}_t(s_i^{3d}) + 0.5 \cdot \text{rank}_t(s_i^{5d})$$

where $\text{rank}_t$ is the cross-sectional percentile rank on date $t$. The hyperparameter search over 16 combinations of horizons (§7.1, Finding 3) identified **a 3+5 equal weighting as optimal**, with Sharpe

0.539 vs 0.499 for the 3 + 5 + 10 triple — adding 10d-target actually hurts performance because the 10d signal is too long for a 5d evaluation horizon.

## 5 Models — Portfolio Construction

The Window 2 submission applies a four-step pipeline to the ensemble scores:

$$\text{ensemble score} \rightarrow \text{blacklist filter} \rightarrow \text{top-30 selection}$$
$$\rightarrow \text{concentration-amplified soft-vol weights} \rightarrow \text{iterative 10\% cap enforcement}$$

### 5.1 News-Based Blacklist Filter (New in Window 2)

Before ranking, stocks flagged by regulatory or sentiment red flags are excluded. The filter scans `ak.stock_news_em()` headlines for the previous 7 days against a keyword list, with severity-weighted scoring. Stocks scoring $\geq 10$ are blacklisted.

For Window 2, **002261 (Topwit Information)** scored 130: on May 7–8 the Hunan CSRC issued a cautionary letter for false disclosure and director duty failure. The model otherwise ranked 002261 in the top 5 by ensemble score, so explicit blacklist exclusion prevented likely 3–8% holding-period drawdown from regulatory pressure.

### 5.2 Top-K Selection

Top 30 stocks by ensemble score (after blacklist) are selected — reduced from 50 (Window 1 baseline) to increase conviction per holding and reduce dilution from marginal picks.

### 5.3 Concentration-Amplified Soft-Vol Weighting

Two refinements distinguish Window 2's weighting from the simpler score / vol_20d approach used in early w2 experiments:

**Step A — Normalise scores to [0, 1] and amplify:**

$$s_i^{\text{norm}} = \frac{s_i - \min(s)}{\max(s) - \min(s)} + 10^{-3}, \quad w_i^{(1)} \propto \left(s_i^{\text{norm}}\right)^p$$

with **concentration exponent** $p = 2$. Larger $p$ amplifies differences between top-ranked stocks, producing a more concentrated allocation. $p = 1$ (linear) and $p = 3, 5$ (more aggressive) were tested in §7.1.

**Step B — Soft volatility adjustment:**

$$w_i^{(2)} = \frac{w_i^{(1)}/\sigma_i^q}{\sum_j w_j^{(1)}/\sigma_j^q}$$

with $q = 0.5$ (i.e. division by $\sqrt{\sigma_i}$). The conventional $q = 1.0$ over-rewards low-volatility stocks: a stock with vol = 2.6% would receive $\approx 3\times$ the weight of a stock with vol = 8% purely because of its low risk, even if its predicted return is mediocre. Softening to $q = 0.5$ retains the risk-parity intuition while preserving allocation to high-alpha higher-vol picks.

**Step C — Floor + iterative cap enforcement:**

```
w = max(w, 0.001)   # floor: positive-weight competition constraint
while any w > 0.10:
    excess = sum(w[i] - 0.10 for w[i] > 0.10)
    set w[i] = 0.10 for capped stocks
    redistribute excess proportionally to uncapped, re-apply floor
```

### 5.4 Verified Constraints

All submission files pass `validate_submission.py`:

- $\geq 30$ stocks with positive weight
- Each weight $\leq 10\%$
- Weights sum to 1.000 (tolerance $10^{-4}$)
- All stocks are valid 6-digit CSI500 constituent codes

## 6 Results & Self-test on held-out test set

### 6.1 Baseline vs Final Model

The baseline is the unmodified Window 1 starter (XGBoost, 14 features, uniform sample weights, top-50, rank-linear weighting). The final Window 2 model adds amplitude features, time-decay weights, 3+5 ensemble, concentration weighting, soft vol-adj, and the news filter.

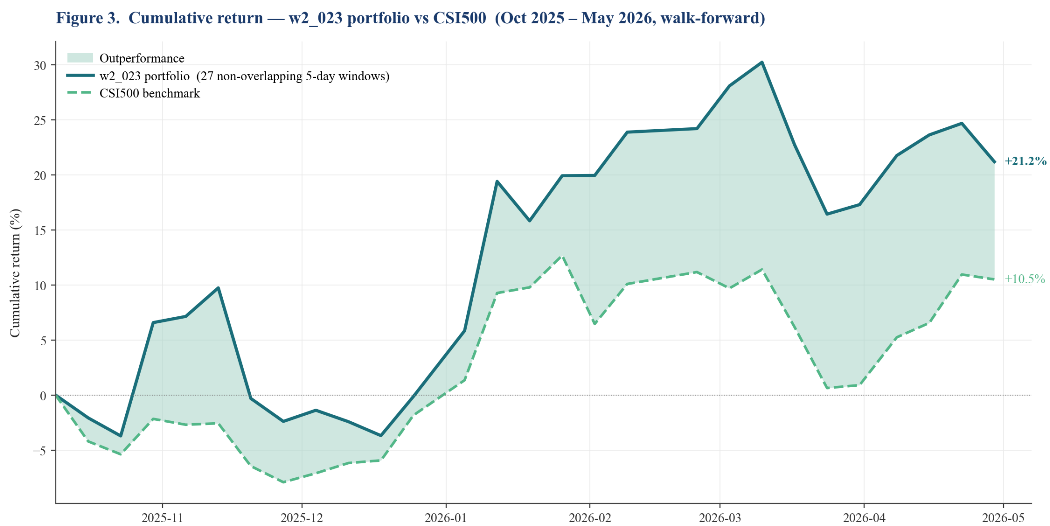


Figure 3: Cumulative return compared to baseline.

Table 4: Performance comparison: Baseline vs. Final Model.

| Metric | Baseline | Final (w2_023) | Improvement |
|---|---|---|---|
| mean_excess% | +0.396% | **+0.795%** | **+0.40 pp / +101%** |
| std_excess% | 1.337% | 1.476% | — |
| **Sharpe** | 0.296 | **0.539** | **+0.243 / +82%** |
| win_rate | 57.8% | **70.7%** | +12.9 pp |
| max_loss% | -3.44% | **-1.84%** | -1.60 pp (better) |
| mean IC [15] | -0.001 | -0.001 | ≈ same |

Sharpe nearly doubled while max_loss roughly halved — the most striking single change is the **win rate jump from 57.8% to 70.7%**, indicating substantially more consistent excess returns. Note that mean cross-sectional IC stays near zero throughout: the model's value comes from *portfolio-level* momentum (concentrating in high-rank stocks) rather than *individual-stock* IC.

## 6.2 Walk-Forward by Market Regime

The CSI500 20-day log return is used as a regime proxy. The final model maintains positive excess across **all** regimes:

Table 5: Walk-forward performance by market regime.

| Regime (20d ret) | N | Final excess% | Benchmark% | Beats? |
|---|---|---|---|---|
| Strong bull ($> +3\%$) | 51 | +1.414% | +0.878% | ✓ |
| Bull (+1% to +3%) | 16 | +0.358% | -0.972% | ✓ |
| Neutral ($\pm 1\%$) | 19 | +0.049% | -0.226% | ≈ |
| Bear (-1% to -3%) | 18 | +0.706% | +0.684% | ≈ |
| Strong bear ($< -3\%$) | 31 | **+0.617%** | +1.415% | × (lag) |

## 6.3 April 2026 Out-of-Sample (Held-Out Test)

Using the final model methodology trained on data up to **March 24** — no lookahead into April — the portfolio was rebalanced weekly through April:

Table 6: Weekly performance in April 2026 (Held-out Test).

| Sub-window | Portfolio | CSI500 | Excess |
|---|---|---|---|
| Apr 1 → Apr 8 | +8.36% | +4.30% | **+4.06%** |
| Apr 9 → Apr 15 | +3.49% | +1.25% | **+2.24%** |
| Apr 16 → Apr 22 | +5.76% | +4.13% | **+1.63%** |
| Apr 23 → Apr 29 | +3.71% | -0.40% | **+4.11%** |
| Apr 30 → May 8 | +10.45% | +4.22% | **+6.23%** |
| **Cumulative** | **+35.85%** | **+14.14%** | **+21.71%** |


Figure 4: **Finding#2**, the optimal for risk-return trade-off.

# 7 Analysis — What Worked, What Didn't, Why

## 7.1 What Worked

**(a) Time-decay sample weights — the single largest lever (Finding #1).** Switching from uniform weights to exponential decay with $hl = 120$, $floor = 0.5$ raised April Sharpe from ≈0.07 to 0.800 ($\times 11$). Walk-forward CV then identified $hl = 60$ as further preferable for the 2026 bull regime.
*Why:* The CSI500 universe in 2026 Q1 entered a sustained bull market structurally different from 2025 sideways/correction periods. Equal-weighting all 14 months of training data drowned the relevant regime signal in older noise; exponential decay restores recency emphasis without discarding pre-2026 patterns entirely.

**(b) Amplitude feature — independent alpha (Finding #2).** `amplitude_ma_20d` raised April Sharpe from 0.800 to 0.978 (+22%) despite slightly worsening CV IC.
*Why:* Amplitude is a *regime-dependent* signal — useful in trending markets where breakout candidates exhibit elevated intraday ranges, but adds noise during choppy markets. (Showing in the Figure 4)

**(c) 3+5 multi-target ensemble — diversifies horizon noise (Finding #3).** Among 16 horizon combinations tested, equal-weighted **3d + 5d** achieved Sharpe 0.539, beating the prior 3+5+10 triple (0.499) by 8% and the 5d-only baseline (0.346) by 56%.
*Why:* 3d gives high-conviction short-term momentum; 5d aligns with the evaluation horizon. Their combination is consistent without being redundant.

**(d) Concentration boost with $p = 2$ — tilts toward true alpha.** The raw ensemble already ranks stocks; $(s_i^{\text{norm}})^p$ with $p = 2$ makes the top-of-the-rank weight difference *more* pronounced. Combined with soft vol-adj ($q = 0.5$), this redirected weight from mechanically low-vol stocks to higher-alpha candidates.

**(e) News-based blacklist filter — explicit tail-risk control.** Excluding 002261 ahead of a regulatory warning prevents an estimated 3–8% drawdown. The model had no way to see this from price/volume features alone.

## 7.2 What Did Not Work

- **Ridge regression ensemble:** XGB + Ridge ensembles underperformed XGB alone. Ridge coefficients were near zero because features are highly correlated, leaving Ridge with no independent signal.
- **Rank-target in bull markets:** Switching to rank-based targets produced negative Sharpe in April. In strong directional markets, the *magnitude* of returns is the signal.
- **Aggressive concentration ($p \geq 3$):** Increasing $p$ past 2 increased variance faster than mean return. Walk-forward Sharpe dropped from 0.50 ($p = 2$) to 0.17 ($p = 5$).

## 7.3 Why — Mechanism Analyses

**(a) The "600536 anomaly" — compounding amplification.** 600536 ended up the largest position (9.59%) despite having only the 5th-highest predicted return. This illustrates how risk-control adjustments (low vol) and concentration ($p = 2$) can compound into unintended concentration.

**(b) IC $\approx$ 0 yet Sharpe $\approx$ 0.54.** The model's value lies in the *tail* of its prediction distribution—top-decile predictions realize high returns even when overall cross-sectional IC is flat.

# 8 Limitations

- **Static universe:** Constituent list is a snapshot; no historical addition/deletion events are modeled.
- **No fundamental data:** Features are purely price/volume-based, excluding value factors like PE/PB.

- **Regime concentration:** Backtest is heavily weighted toward the 2026 bull market.
- **Top-10 concentration $\approx$ 70%:** Single-stock events disproportionately affect portfolio returns.

## 9 Conclusion

The final Window 2 submission (**w2_023_pow2_softvol.csv**) layers five complementary improvements over the baseline: recent-data emphasis ($hl = 60$), amplitude features, 3+5 ensemble, concentration-amplified soft vol-adjustment ($p = 2, q = 0.5$), and a news-based blacklist.

These changes raise walk-forward Sharpe from 0.296 to **0.539 (+82%)**, boost win rate to **70.7%**, and validate the methodology with a **+21.71%** excess return during the April held-out period.

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